

WAH4PCE: An AI-Driven Interoperability Bridge for iHOMIS

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Executive Summary

The realization of the Universal Health Care (UHC) Act across the Philippines is critically hindered by isolated "data islands," where clinical records remain locked within fragmented health software systems. While Wireless Access for Health (WAH) facilitates modern HL7 FHIR-compliant exchanges among its partnered clinics, cross-facility communication completely breaks down when sharing records externally with Department of Health (DOH) facilities running the Integrated Hospital Operations and Management Information System (iHOMIS). This format mismatch forces healthcare personnel to rely on slow, error-prone manual re-encoding, increases the risk of data pollution, and exposes facilities to Data Privacy Act (RA 10173) violations during cross-network patient referrals.

To address this critical gap, WAH4PCE (Wireless Access for Health for Interoperability External) deploys the ADAPT (Automated Data Processing and Transformation) middleware engine as a localized API gateway. The system employs a robust Medallion Data Architecture to stage and process information securely through Bronze (raw health data), Silver (validated and cleansed data), and Gold (translated, interoperable data) layers.

- **AI-First Translation Engine:** The core translation infrastructure relies on an Artificial Intelligence engine to achieve semantic interoperability, handling the complex mapping of medical context between WAH (FHIR R4) and DOH iHOMIS (HL7 v2) schemas. To ensure absolute data privacy, the final vision utilizes a localized, on-premises AI server, while this Proof of Concept (PoC) utilizes Google Gemini via MCP to validate the mapping logic. A deterministic, precise algorithm is strictly utilized as a fallback safety net to ensure structural, syntactic interoperability.
- **Automated Privacy Gatekeeping:** The middleware proactively isolates missing mandatory fields (such as PhilHealth numbers) to prevent data pollution. It features

an automated privacy gatekeeper that instantly halts and quarantines unauthorized transfers lacking verified digital patient consent, valid consent flags, and proper validity periods.

- **High-Performance, Scalable Architecture:** Built on a modern tech stack (Golang, MongoDB, Next.js), the system utilizes an intelligent mapping layer that processes in-memory payloads without permanently retaining confidential patient data.

By establishing this secure, AI-driven API prototype, defined by encrypted local network routing, strict mutual API key authentication, and automated consent validation, WAH4PCE provides a proven, scalable model to eliminate data silos and safeguard patient rights without overextending into an unfeasible nationwide infrastructure development.

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Introduction

In many cases, healthcare suffers from the way patient information is stored and shared. As each hospital or clinic operates its own independent computer system, transferring patient records between healthcare facilities remains a significant challenge, resulting in isolated "data islands". Within the WAH network, partnered clinics and hospitals can easily share data because their systems use the HL7 Fast Healthcare Interoperability Resources (FHIR) R4 standard. However, a major barrier occurs when a WAH-partnered hospital needs to send patient data externally to a government hospital using iHOMIS, as the two systems utilize completely different data formats. This incompatibility results in slow care and a heavy reliance on manual, pen-and-paper processes.

While creating a basic data connection might achieve syntactic interoperability, the ability to exchange structured data, the true challenge lies in achieving semantic interoperability. Semantic interoperability ensures the receiving hospital actually understands the clinical meaning of the data without context loss. The Universal Health Care (UHC) Act mandates the implementation of interoperable information systems, but the development of a national infrastructure is currently on hold, creating a critical need for local organizations to establish their own platforms. The previous WAH4PC development batch successfully established a baseline interoperability server, but its capabilities were strictly limited to internal communication within the WAH ecosystem. As the direct technical inheritor of this baseline, WAH4PCE upgrades the infrastructure by introducing the Automated Data Processing and Transformation (ADAPT) engine. Designed as a highly focused, localized Proof of Concept (PoC), WAH4PCE acts as a functional API bridge rather than attempting to construct a massive nationwide architecture.

To ensure data integrity, the system implements a Medallion Data Architecture. Incoming health records are ingested as raw payloads (Bronze layer), cleansed and validated for completeness (Silver layer), and structured into interoperable target formats (Gold layer). Crucially, the primary translation engine powering this transformation is an

Artificial Intelligence (AI) engine designed to read and translate medical meaning. For this Proof of Concept, Google Gemini via a Model Context Protocol (MCP) server is employed strictly to validate the semantic mapping logic and demonstrate feasibility. However, to guarantee absolute data privacy and security, the final production deployment will utilize a localized, on-premises AI server. To support this AI-first approach, a deterministic, precise mapping algorithm acts as a strict fallback safety net to process standard syntactic edge cases if the AI encounters downtime.

Additionally, to adhere to the right to data portability under the Data Privacy Act of 2012 (RA 10173), the system operates as an automated legal gatekeeper. It scans incoming records to verify the presence of explicit legal consent, automatically blocking and quarantining the transfer if the consent is missing or invalid. The finalized API gateway will be exposed to external partners, such as DOH iHOMIS, alongside complete instructions and standard documentation to ensure clear, formal, and strictly governed integration guidelines.

Statement of the Problem

The primary problem is the lack of an automated, working system that can securely translate and share patient records between local WAH clinics and outside government hospitals (iHOMIS).

Specifically, this study aims to answer the following problems:

1. **Missing Outside Connections:** The current WAH4PC system is built only for internal use; it lacks the encrypted connection points and standard API manuals needed to link securely with external Department of Health (DOH) systems.
2. **Poor Data Quality Control:** The system lacks a step-by-step filtering process to catch incomplete patient records (such as those missing PhilHealth IDs) before they are shared, which risks spreading bad or missing data across hospital networks.
3. **Syntactic Translation Bottlenecks:** Translating structured patient data between the WAH format (FHIR R4) and the iHOMIS format (HL7 v2) currently relies on slow manual typing. There is a critical need for an automated algorithmic engine capable of handling baseline syntactic interoperability by exactly matching structural schemas.
4. **The Semantic Interoperability Gap:** A strictly rule-based algorithmic program is syntactically rigid; it cannot process unstructured data, non-standard medical abbreviations (e.g., typing "pt has SOB" instead of a coded "Shortness of Breath"), or mismatched clinical terms. This requires an intelligent, AI-driven engine to achieve semantic interoperability, reading and translating the actual clinical meaning of confusing or unmapped edge cases.

5. **Privacy Risks:** The current setup lacks an automated privacy checker, creating a serious legal risk of sending sensitive patient records to outside hospitals without first verifying the patient's digital consent form, consent flag, and validity period.

Objectives

The primary objective of this project is to design and implement an AI-driven API bridge that securely translates patient records between the internal WAH ecosystem and external iHOMIS Implementing Hospitals.

Specifically, this project aims to:

1. **To connect the internal system to DOH iHOMIS:** Upgrade the current server into a secure gateway, utilizing encrypted local network routing and mutual API key authentication, and provide simple, standardized API documentation so external government hospitals can easily connect to it.
2. **To ensure data quality using a step-by-step filter:** Create a three-step data checking process (Medallion Architecture's Bronze, Silver, and Gold layers) that automatically cleans the records and blocks incomplete forms before they are sent.
3. **To automate syntactic translation using strict rules:** Develop a fast, exact computer algorithm that acts as a strict safety net to automatically map the structural schemas between the WAH and iHOMIS formats, ensuring baseline syntactic interoperability.
4. **To achieve semantic interoperability using an AI-First engine:** Build an intelligent primary translation engine to map complex health records and understand unstructured clinical context, using Google Gemini AI as a Proof of Concept to simulate the logic for a future localized, privacy-secure AI server.
5. **To automatically enforce data privacy laws:** Build a digital legal gatekeeper that scans every medical record and immediately blocks the transfer if the patient's explicit consent form, consent flag, or consent validity period is missing or invalid.

Significance of the Project

This project offers practical, immediate benefits by improving how health data is shared securely within a realistic timeframe. The specific groups that will benefit from this project are:

- 1. For WAH and DOH:** It provides a working, localized API bridge Proof of Concept (PoC). This proves that local systems and national government hospitals can successfully communicate using an AI-driven engine with algorithmic fallbacks today, without having to wait for a massive national infrastructure to be built.
- 2. For Healthcare Staff:** It eliminates the tedious task of double data entry. Nurses and IT staff will no longer need to manually retype medical records or fix formatting errors when referring patients from a WAH clinic to an iHOMIS hospital, saving valuable time and reducing human error.
- 3. For Patients:** It guarantees their legal right to privacy through automated consent checks. Furthermore, it ensures their complete and accurate medical history travels securely and instantly alongside them during hospital transfers, resulting in faster and safer care.

Scope and Limitations

Scope:

1. **API Gateway Proof of Concept & Concurrent Processing:** The project upgrades the inherited baseline server into a centralized, external-facing API gateway designed to be completed within a 6-to-12-month timeframe. It utilizes modern concurrent processing (Golang) to efficiently handle simultaneous data requests from multiple independent hospital connections.
2. **External System Connectivity via API:** The scope establishes a bidirectional communication pathway tailored specifically for iHOMIS, enabling WAH-partnered hospitals to send records outward and receive incoming clinical data. This includes providing formal, standard API documentation for seamless external integration.
3. **Medallion Architecture Data Validation & Consent Verification:** The system implements a structured data staging pipeline (Bronze, Silver, Gold). Automated checkers at the Silver layer proactively isolate structurally incomplete records (e.g., missing PhilHealth numbers). It also acts as a digital gatekeeper, automatically blocking the transfer of records lacking an explicit patient consent flag, form, or validity to ensure strict adherence to the Data Privacy Act.
4. **AI-First Format Translation:** Replacing manual data entry, the primary translation engine relies on an Artificial Intelligence model to intelligently map complex clinical records and unusual data, ensuring semantic interoperability between the WAH network (FHIR R4) and external government systems (HL7 v2).

5. **Algorithmic Fallback Safety Net:** A precise, deterministic algorithm is integrated exclusively as a backup mechanism to process records and maintain syntactic interoperability if the primary AI engine encounters unmapped edge cases or downtime, keeping processing highly reliable.

Limitations:

1. **Centralized Architecture Constraint:** Following advisory constraints, the project scope is limited to a centralized database staging approach. It does not attempt to build a decentralized or federated hospital-to-hospital routing network.
2. **Reliance on Established Standards:** The system does not create new data formats or health standard structures; it operates strictly within the pre-defined boundaries of the established PH Core and iHOMIS data schemas.
3. **Endpoint Availability Dependencies:** The confirmation of functional bidirectional data exchange is completely dependent on the availability and simulated readiness of external hospital and clinic system endpoints managed by separate development teams.
4. **AI Proof of Concept & Localized Server Vision:** To validate the AI mapping logic without heavy up-front server infrastructure expenditures, this prototype relies on cloud-based processing (Google Gemini) strictly as a PoC. However, the final operational deployment will shift to a localized, on-premises AI server managed by WAH to guarantee absolute data privacy.
5. **Simulated Testing Boundaries:** Due to strict privacy laws, the project is explicitly barred from accessing or using live, real-world public health databases. System validation is limited to testing environments utilizing synthetically generated health data within simulated hospital endpoints.

6. **Clinical Truth Exclusions:** The system's checking tools are strictly limited to enforcing structural format accuracy, mapping logic, and legal consent flags; the architecture cannot evaluate, edit, or verify the actual clinical correctness or medical truth of the information submitted by a facility.

Review of Related Literature / Systems

The landscape of healthcare is undergoing significant metamorphosis driven by digital transformation. This evolution, while promising enhanced efficiency and patient care, hinges on various factors, including technology acceptance and the effective implementation of digital tools.

One of the pioneering studies that shed light on this shift in the industry is the work of Nassr et al. [12]. The authors investigate various aspects of how healthcare professionals and patients are adopting new technologies. Specifically, the authors investigate the factors that make healthcare professionals more or less likely to adopt these technologies.

Another study that investigates this same topic but with a focus on the local area is by [25], who investigate the current state of the Electronic Health Record (EHR) system in the Philippines. The authors reveal valuable insights into the current state of the EHR system in the Philippines. Furthermore, because there are some disconnected EHR systems within the Philippines, there is a recognized need for a system that can provide a standard system for the data to be exchanged between these disconnected systems.

The digital tools can be further seen in the work of [21], who discuss the potential of artificial intelligence to improve health outcomes through earlier and more accurate diagnoses.

The benefits and challenges of EHRs are discussed by [19]. The authors explain how these systems can lead to improved health information management and health care professionals' workflows, but also present issues with interoperability between different health systems. Most health systems utilize slightly different structures of their internal data from others in the industry, preventing interoperability between these health records databases.

To overcome these interoperability challenges, the healthcare industry has used the HL7 FHIR standard [1]. HL7 FHIR allows healthcare data to be represented in a standard format using modular resources. Using this standard, systems can exchange data with semantic interoperability. FHIR requires full mapping from the existing databases to the FHIR standard to enable interoperability between these two systems. Several research studies have explored how to create interoperable and reusable health data using the FHIR standard [20]. Additionally, there are several studies exploring the semantic issues of FHIR and implementing the FHIR standard within EHR systems [3]. These solutions will eliminate data silos and enable better communication between systems.

Finally, the broader implications for healthcare quality and safety are discussed by Alshurideh et al. [2]. Their work highlights how digital transformation improves service delivery, particularly when systems are integrated and capable of sharing accurate data. The effectiveness of these improvements depends on the ability of systems to communicate reliably, further reinforcing the importance of standardized interoperability frameworks such as FHIR.

Through reviewing the literature, it is evident that there is both a global and local movement toward the digitization of healthcare. The literature discusses, for instance, the acceptance of technology in healthcare [12], the challenges of implementing Electronic Health Records in healthcare facilities [8], the role that advanced technologies play in improving healthcare outcomes [21], and the impact of digitization on healthcare [2]. Furthermore, systems like the Community Health Information Tracking System (CHITS) and iHOMIS have allowed for the digitization of certain processes within the Philippines [10, 13, 14, 18] but also demonstrate the limitations of such systems implemented alone. Thus, the literature indicates a need for a system that provides interoperability for the country's healthcare systems. Such interoperability could be achieved through the implementation of a system that maps each healthcare system to the FHIR standard, which would guarantee

compliance with the UHC law [17, 18] and lead to the integration of the entire Philippine healthcare system.

According to research by [11], the biggest obstacle to digital transformation in healthcare is the existence of so-called data islands. Even as medical records are increasingly going digital, the data are still confined to heterogeneous Information Technology (IT) systems that lack a common language. This fragmentation does not allow for providing coordinated care and narrows the utility of clinical decision-support tools [11].

The researchers refer to FHIR as the most promising solution to this issue. Unlike the previous messaging standards, which were complex and rigid, FHIR leverages the current web technologies i.e. RESTful (Representational State Transfer) API and modular resources [11]. They consist of standardized building blocks, including Patient, Medication, and Procedure, which are shared between different systems in order to share certain bits of information efficiently. Flexibility of FHIR is highlighted in the literature, with the mention of the fact that it can be applied to various platforms, such as mobile health apps and cloud systems [11].

Nan and Xu [11] formulate three areas of critical importance on the basis of a systematic review of 85 scholarly articles:

- **Data Standardization and Mapping:** A sizeable portion of the literature is focused on the transformation process, i.e. how to convert the current legacy data, e.g. CDA or HL7 v2, to FHIR-conformant data. This is critical in making sure that information from various time periods and systems can be successfully merged [11].
- **Data Management and Storage:** The review explains the architecture of FHIR servers and emphasizes the need to have robust architecture that can address large-scale "Big Data" and maintain high rates of security and privacy controls to protect confidential patient information [11].

- Clinical Services Data Interoperation: This paper goes beyond mere technical exchange to emphasize "semantic interoperability." This has been determined as the ability of systems to not only communicate its content but also the meaning and react to it, which will allow the provision of higher-level services like real-time patient tracking and research across institutions [11].

The proposal of a General FHIR Practice Guideline (FHIR PG) is one of the important contributions of this research. In their argument, [11] argue that standardization is not a technical undertaking but a team effort. Their model defines the roles of different groups of stakeholders to be successful: to prevent the non-compatibility of different versions of FHIR, the standardizers need to create particular Implementing Guides; the system provider needs to create the infrastructure and servers to store such data; and the service provider (app developers) needs to create user-facing tools to transform this data into improved clinical outcomes [11].

The transition from manual paper-based reporting to EHRs in the Philippines, led by initiatives like WAH, necessitates a theoretical framework for seamless data exchange. According to [24], the FHIR standard serves as this framework by utilizing modern web technologies such as RESTful APIs and JavaScript Object Notation (JSON) to ensure that health data adheres to the FAIR Principles (Findable, Accessible, Interoperable, and Reusable). In the context of the Philippines' Universal Healthcare Act, the literature emphasizes that "data liquidity", the ability of data to move securely and meaningfully between systems, is the next frontier for digital health. [24] argue that FHIR's "resource" building blocks allow for the modular development of health systems, which aligns with WAH's open-source philosophy. However, the literature also warns of significant bottlenecks, including the "versioning" of FHIR resources and the legal complexities of cross-institutional data sharing, which remain primary challenges for the Philippines as it seeks to harmonize fragmented local health information into a unified national repository.

The landscape of digital health in the Philippines is governed by Department of Health and Philippine Health Insurance Corporation (DOH-PHIC) Joint Administrative Order (JAO) No. 2021-0002, which institutionalizes the Mandatory Adoption and Use of National Health Data Standards for Interoperability. This policy serves as the regulatory backbone for the UHC Act, recognizing that "seamless and interoperable health services" are essential for evidence-based sectoral planning. According to the JAO, the current inconsistent adoption of data standards prevents healthcare providers from sharing medical records for continuing care, a critical challenge for decentralized systems like WAH. By mandating a "common data language" through the National Health Data Dictionary (NHDD), the order ensures that health data, whether from a rural health unit or a tertiary hospital, can be analyzed for national program planning and decision-making.

Furthermore, the JAO defines interoperability through a structured set of National Health Data Standards, which include specific data elements, official definitions, and standardized data types (e.g., text, numeric, and date formats). This policy framework supports the "FAIR" principles (Findable, Accessible, Interoperable, and Reusable) discussed by [24] by providing the legal requirement for systems to utilize authoritative terminology services. For local health initiatives, compliance with these standards is now a prerequisite for licensing and accreditation, effectively making technical interoperability a mandatory operational standard rather than an optional technical feature [9].

The global trajectory of healthcare informatics is increasingly defined by the transition from localized data digitization to the pursuit of semantic interoperability. Semantic interoperability refers to the ability of different computer systems to exchange information with a shared, unambiguous meaning so that the receiving platform can process the data accurately. As highlighted by [5], while the post-Health Information Technology for Economic and Clinical Health (HITECH) Act era has catalyzed the widespread adoption of EHRs, the clinical utility of this data remains hindered by fragmented architectures and variable implementation methods. The HITECH Act is a legislative framework enacted to accelerate

the adoption of digital health platforms, while Electronic Health Records are secure, digital versions of a patient's medical charts. Furthermore, fragmented architectures mean that different software systems are built on completely disconnected technical foundations that prevent them from communicating smoothly. The study posits that the lack of standardized data formats across institutions creates significant barriers to real-time clinical analysis and multi-institutional research. To mitigate these "data silos," the FHIR standard has emerged as the premier technical framework for achieving a unified data language. Data silos are isolated pockets of information trapped inside a single system and cut off from others, whereas the FHIR standard is a universally recognized set of rules and formats designed to streamline the exchange of electronic health data. By utilizing a "resource-based" approach, FHIR provides the necessary modularity to transform disparate clinical narratives into structured, machine-readable formats, supporting the baseline so that health data is not merely digitized but truly interoperable [5]. A resource-based approach refers to a software strategy that breaks down medical data into small, interchangeable units, such as patient demographics or laboratory results, that can be easily shared and reassembled by separate systems.

Furthermore, the research underscores that the adoption of FHIR is a foundational requirement for the advancement of Machine Learning-enabled Clinical Information Systems (ML-CISs). By evaluating 39 distinct systems, the study categorizes these technological advancements into three functional domains: clinical decision support systems, data management and analytic platforms, and auxiliary modules or APIs. This taxonomy demonstrates that the "FAIR" principles (Findable, Accessible, Interoperable, and Reusable) are realized through the automation of data mapping and the reduction of manual pre-processing [5]. Despite these advancements, many intelligent systems continue to face critical challenges regarding external validation and seamless integration into existing clinical workflows. Consequently, the findings suggest that the future of digital health lies in the

development of secure, scalable, and interinstitutional data exchange protocols that can accommodate diverse electronic health record platforms [5].

The transition toward seamless data exchange between highly disparate systems requires robust and scalable architectural frameworks. According to Jayaraman and Singh [25], employing a microservices architecture is the established industry standard for achieving cross-industry interoperability. This type of architecture breaks down a large software system into small, independent services that handle specific tasks individually. The authors emphasize the critical necessity of utilizing specialized "adapter" or "gateway" microservices that manage protocol translation and format conversion without requiring disruptive modifications to existing legacy systems [25]. In plain terms, these components act as dedicated translators that sit between two different systems to convert data without altering the original software. This principle provides a strong theoretical and practical validation for the WAH4PCE project, specifically justifying the development of the Automated Data Processing and Transformation (ADAPT) engine as an optimization upgrade to the operational WAH4PC baseline to resolve traffic management and operational visibility constraints. By functioning as an independent, intelligent middleman, the ADAPT engine translates medical records between the WAH network's FHIR R4 standard and the iHOMIS format, directly mirroring the adapter-pattern best practices outlined in the study [25]. The adapter pattern refers to a software design strategy where an intermediary tool standardizes communication between incompatible interfaces. Furthermore, Jayaraman and Singh [25] strongly advocate for decentralized data management and strict API boundary security to mitigate the risks of single points of failure and help protect data confidentiality during transit. Decentralized data management means that information remains safely stored locally at each individual hospital rather than being gathered into one massive, vulnerable central server. This proves the profound relevance and necessity of the project's federated network architecture, its interactive frontend dashboard, and its AI-driven legal gatekeeper. A federated network architecture is a digital setup where separate, independent local servers

safely connect and share data directly with one another. By applying the decentralized and stateless processing principles established in the literature [25], which dictate that patient information is handled strictly in-memory and is completely wiped from the middleman's system the moment the transfer is complete, the WAH4PCE system supports the secure, efficient achievement of healthcare interoperability in strict compliance with data privacy regulations.

The objective to automate health data interoperability and reduce reliance on manual data entry is not solely a technical upgrade, but a necessary step to ensure patient safety and data integrity. According to Abraham et al. [26], patient acceptance of advanced health information systems is heavily influenced by the perceived reliability of the technology; patients frequently express apprehension regarding data inaccuracies, system malfunctions, and human errors when their sensitive information is transferred across networks. By automatically translating medical records between WAH and iHOMIS formats, the WAH4PCE system directly mitigates these socio-technical concerns. Eliminating the need for manual re-encoding significantly reduces the risk of human-introduced transcription errors, often referred to as data pollution, ensuring that a patient's medical history remains accurate as it moves between facilities. In this context, the automated translation engine functions as the type of built-in "safeguard system" that Abraham et al. [26] identify as crucial for establishing patient trust and ensuring the secure, reliable transfer of health data.

The necessity of ensuring data integrity and strict schema compliance prior to network transmission extends beyond operational efficiency; it is a fundamental requirement for maintaining data security in modern medical environments. According to Yadav et al. [27], the deployment of artificial intelligence and digital health repositories requires "fool-proof" prerequisites to protect sensitive patient information from the immense vulnerabilities inherent in massive data transfers. When health data is exchanged between disparate systems without stringent formatting checks, the introduction of incomplete, unverified, or anomalous information, commonly referred to as data pollution, can severely compromise

the security and privacy protocols of the entire network. By fulfilling the objective to integrate an automated validation tool that verifies translated records for consistency and missing mandatory fields, the WAH4PCE system establishes the rigorous data governance necessary to safeguard health information. Yadav et al. [27] emphasize that navigating the ethical and legal constraints of healthcare data sharing requires strict oversight. Therefore, by ensuring that only clean, structurally compliant, and fully verified data is allowed to transmit between WAH and iHOMIS, the system proactively blocks corrupted or incomplete payloads that could lead to privacy breaches or database corruption [27].

Data Governance through Medallion Architecture The transition toward seamless inter-facility data exchange requires robust data quality governance to prevent system-wide contamination. In modern data engineering, the Medallion Architecture offers a layered approach to incrementally refine data and handle quality control [28]. This method simplifies data processing by organizing it into three distinct levels: bronze, silver, and gold [28]. According to research on data engineering frameworks, the bronze level is responsible for taking in raw data and securely storing it [28]. The silver layer then focuses on cleaning, checking, and standardizing the data to fix quality issues and ensure it meets strict business rules [28]. Finally, the gold layer provides the fully processed and structured data ready for system consumption or business assessment [28]. By applying this framework to the WAH4PCE gateway, malformed payloads, such as records missing mandatory PhilHealth identifiers, can be isolated at the silver level before reaching the final translation phase. This step-by-step refinement provides structured quality management and ensures that only compliant data is transmitted.

Syntactic vs. Semantic Interoperability and the AI Imperative Interoperability in healthcare information systems is traditionally categorized into distinct levels, most notably syntactic and semantic interoperability. Syntactic interoperability refers to the ability of systems to exchange structured data predictably, relying on standard formats like JSON or XML. While deterministic, rule-based algorithms excel at achieving this structural alignment,

they are fundamentally limited in their scope. True data liquidity requires semantic interoperability, the ability of disparate systems to not only exchange data but also preserve, interpret, and understand its underlying clinical meaning without context loss or manual clarification [30]. Real-world legacy systems, such as iHOMIS, frequently utilize unstructured clinical narratives, non-standard abbreviations, and custom data segments that break rigid algorithmic rules. Recent research highlights that Large Language Models (LLMs) can effectively bridge this semantic gap by acting as a cognitive interface, interpreting unstructured clinical text and translating it into standardized HL7 FHIR formats [31]. By deploying an AI-driven primary engine for semantic understanding, backed by a deterministic algorithm for syntactic mapping, the WAH4PCE system successfully mitigates the brittleness of traditional data exchange, ensuring accurate and meaningful interoperability.

Algorithmic Management and LLM Governance While the integration of Artificial Intelligence offers powerful capabilities for handling complex data edge cases, it also presents significant operational and security risks if deployed without strict constraints. A recent study [29] explores the risks and necessary controls of Large Language Model (LLM) integration within organizational systems. The research emphasizes that while LLMs can effectively bridge structural gaps by acting as a cognitive interface, their integration introduces severe risks such as "hallucinated rationales," "bias amplification," and "automation dependence" [29]. To safely harness AI, the study proposes a governance-oriented architecture that combines a rigid "algorithmic decision core" with a strict "verification and governance layer" that enforces policy constraints, audit trails, and oversight [29]. This governance model provides profound theoretical validation for the WAH4PCE system's architectural setup. By establishing an AI-first approach as the primary semantic translation engine, and restricting the system with an exact, deterministic algorithm serving as a fallback safety net for standard syntactic mapping, the gateway remains technically explainable, legally defensible under data privacy laws, and highly resistant to AI-generated hallucinations. Furthermore, by embedding an automated legal consent

gatekeeper within the Medallion Silver layer, WAH4PCE satisfies the literature's requirement for a robust verification layer [29].

Technical Background

To resolve the communication disconnect between the WAH network and government systems, the proposed solution introduces an intelligent middleware named ADAPT (Automated Data Processing and Transformation). ADAPT acts as a secure centralized API gateway between the two networks. Crucially, this project upgrades the previous WAH4PC baseline system. It transforms the original, internal-only background server into a high-speed, external-facing platform equipped to securely route and intelligently translate health records directly between WAH-partnered clinics and iHOMIS Implementing Hospitals.

Current System

The current baseline architecture, WAH4PC (developed by TCG Trio in Batch 1), was established as a centralized proof-of-concept interoperability server designed to facilitate data exchange within the internal WAH ecosystem (e.g., between WAHTERMELON and local clinic systems).

- **Hardware and Deployment Architecture:** The baseline system operates on a centralized server infrastructure accessed remotely by connected partner facilities. The server hosts the application backend, API routing layers, and local caches (e.g., Redis) to handle incoming requests.
- **Software Technologies:** The current system relies primarily on a Python-based backend utilizing lightweight web frameworks (Flask or Django) alongside standard Python FHIR processing libraries to construct and manipulate HL7 FHIR resources. AI integration in the baseline relies on an external Model Context Protocol (MCP) server managing custom parser tools to transform source EMR schemas into basic FHIR formats.

- **Network and Communication:** The system communicates using standard RESTful APIs over HTTPS. Inter-facility access and remote server interactions are secured primarily through Virtual Private Network (VPN) tunnels connecting participating clinics to the central host.
- **Peopleware:** System operations depend on WAH system administrators, IT support staff, and clinic data encoders who initiate records and manually monitor logs for transaction errors or failed transformations.

Limitations of the Current System

1. **Internal-Only Scope:** The baseline architecture was designed strictly for intra-ecosystem exchange. It lacks the data adapters, secure endpoints, and schema definitions required to communicate with external government platforms like DOH iHOMIS (which utilizes HL7 v2 structures).
2. **Concurrency and Traffic Bottlenecks:** Built primarily in Python for prototyping, the system exhibits performance and concurrency limitations when subjected to heavy, simultaneous data traffic loads across multiple hospital connections.
3. **Lack of Visual Administrative Oversight:** The baseline runs as a background service without a dedicated, interactive web interface, leaving administrators without real-time visibility into queue health, translation metrics, or quarantined payloads.
4. **Lack of Data Governance and Legal Automation:** The system lacks a staged filtering pipeline to catch incomplete records, risking data pollution. Furthermore, privacy validation relies heavily on manual verification, lacking an automated legal gatekeeper to enforce the Data Privacy Act before data transmission.
5. **Syntactic Translation Limitations:** The baseline struggles with semantic interoperability. It relies on basic, rigid parsing that cannot intelligently handle the

unstructured data, medical jargon, or mismatched clinical terms prevalent in legacy systems like iHOMIS without a robust fallback mechanism.

Proposed System

The proposed system, WAH4PCE (WAH4PC External), introduces the ADAPT middleware to directly resolve the traffic, visibility, and external interoperability constraints of the inherited baseline. It bridges WAH facilities (FHIR R4) and government hospital systems (iHOMIS / HL7 v2) while enforcing strict compliance with the Data Privacy Act of 2012 (RA 10173) and the Universal Health Care Act (RA 11223).

- **Software Technologies:**

- **Core Processing Engine (Golang):** The backend routing orchestrator is developed in Go (Golang), leveraging its native concurrency and goroutines to handle high-volume, simultaneous data transfer requests across independent hospital networks with low latency.
- **Medallion Architecture and Legal Gatekeeper:** The system implements a structured data staging pipeline (Bronze, Silver, Gold). At the Silver layer, it features an automated privacy gatekeeper ("Consent-as-Code") that parses incoming payload metadata. It isolates missing mandatory fields (such as PhilHealth IDs) to prevent data pollution, and immediately aborts unauthorized transfers if an explicit patient consent form, flag, or validity period is absent.
- **AI-First Semantic Engine with Algorithmic Fallback:** The system utilizes an Artificial Intelligence engine as the primary translator to achieve semantic interoperability, intelligently mapping complex medical context between FHIR R4 and HL7 v2 schemas. For this Proof of Concept, Google Gemini via MCP is used strictly to simulate the logic for a future privacy-secure, localized AI server. An exact, deterministic mapping algorithm is strictly utilized as a fallback safety net to ensure syntactic interoperability if the AI encounters unmapped edge cases or downtime.

- **Administrative Dashboard (Next.js):** A modern, responsive frontend built with React providing IT administrators with visual control over real-time pipeline metrics, live transaction logs, quarantine inspection tools, and API configuration management.
- **Data Staging (MongoDB):** A flexible, document-oriented database used for centralized in-memory staging and temporary payload caching during transformation, ensuring no raw clinical data is permanently retained post-transmission.

- **Hardware and Deployment Infrastructure:**
 - **Virtual Environment Simulation (Proxmox VE):** To test real-world distributed hospital deployments without costly physical setups, the system is deployed and simulated across isolated virtual machines on local Proxmox VE servers, mimicking distinct hospital nodes.
 - **Compute and Server Resources:** Dedicated server environments hosting the compiled Go binaries, MongoDB instance, and local MCP connectors, utilizing modern multi-core processors and optimized memory allocation for in-memory data processing.
- **Network and Security Technologies:**
 - **Federated/Decentralized Communication Pathway:** Shifts away from a vulnerable single central database hub toward a direct, node-to-node gateway architecture that preserves local data ownership at each hospital.
 - **Transport Encryption & Zero-Trust Security:** Enforces strict Transport Layer Security (TLS 1.2/1.3) across all API endpoints, mutual API key authentication for every requesting node, and operates seamlessly within the encrypted VPN tunnels established by the parallel security framework.
 - **Sanitized Audit Trails:** Stores immutable transaction logs containing only operational metadata (timestamps, source/target IDs, success/failure status) while completely omitting identifiable patient health data.
- **Peopleware:**
 - **WAH4PCE Administrators:** IT and system administrators who oversee the middleware pipeline, monitor system health, manage facility API keys, and review quarantined records via the Next.js dashboard.
 - **Healthcare Providers & Hospital IT Staff:** Clinical personnel and records officers in WAH-partnered clinics and iHOMIS hospitals who experience

automated, accurate record transfers without manual re-encoding or workflow disruption.

Methodology

This project follows a phased, iterative methodology to ensure structured development, continuous stakeholder involvement, and alignment with the goals of upgrading the existing WAH4PC platform. The process is guided by the Software Development Life Cycle (SDLC), adapted to suit the constraints of a 6-to-12-month student-led project. Each phase is described below:

1. Planning and Requirements Gathering The phase began by mapping out the communication disconnect between separate hospital systems and assessing the specific traffic management and concurrency limitations of the operational WAH4PC baseline platform. Regular consultations were conducted with WAH consultants and officials to clarify existing integration issues, discuss automated consent tracking under the Data Privacy Act (RA 10173), and define the scope of the ADAPT middleware. Key deliverables included the statement of the problem, project objectives, technical scope, and the initial system architecture using a centralized API gateway design.

2. Analysis and Documentation This phase focused on creating detailed technical documentation of the system requirements needed to enhance the current platform and support the new translation and data governance features. The development team created use case diagrams, user stories, and a product backlog outlining the processing requirements for the Medallion Data Architecture and the AI-first translation engine. This phase also identified priority features, data flow sequences across the Bronze, Silver, and Gold layers, and key system actors, including the WAH4PCE Administrator, the WAH-partnered Clinic Administrator, and the external iHOMIS system endpoints.

3. Design Phase (Current Phase) The project is currently in the design phase, focusing on developing wireframes for the new administrative dashboard and finalizing the centralized network architecture simulated on local Proxmox VE servers. The visual wireframes map out user-facing configurations, transaction logs, and data-syncing statuses to give operators full

visibility over network traffic and quarantined records. The proposed tech stack was also finalized:

- **Golang:** For high-speed concurrent processing and traffic orchestration.
- **Medallion Pipeline & MongoDB:** For ephemeral, in-memory data staging through raw (Bronze), cleansed (Silver), and translated (Gold) layers without permanently storing sensitive records.
- **AI-First & Fallback Architecture:** An intelligent AI translation engine (using Google Gemini via MCP as a Proof of Concept for a future localized, on-premises AI server) to handle semantic interoperability, paired with a deterministic, rule-based algorithm as a safety net for syntactic fallback.
- **Next.js:** For the visual administrative monitoring dashboard.

4. Development Phase (Upcoming) Once the user interface designs and local server testing environments are finalized, full-scale development will begin. Features will be built in iterative sprints, starting with the core Go routing engine and the Medallion staging pipeline. Development will then focus on integrating the automated legal gatekeeper (verifying patient consent forms, flags, and validity periods at the Silver layer) and configuring the AI semantic translation layer alongside the deterministic fallback algorithm. Testing will be conducted continuously alongside code development to identify schema errors and API connection issues early.

5. Testing and Quality Assurance The development team plans to conduct rigorous testing using functional dummy system endpoints and synthetic patient data to simulate both a WAH-partnered clinic environment and an iHOMIS Implementing Hospital environment, maintaining strict compliance with the Data Privacy Act. Testing will evaluate:

- **Data Cleansing & Privacy Gatekeeping:** Ensuring malformed payloads (e.g., missing PhilHealth IDs) and records lacking explicit patient consent are properly caught and quarantined at the Silver layer.

- **Semantic and Syntactic Mapping:** Measuring translation accuracy from FHIR R4 to HL7 v2, verifying the AI engine's ability to handle complex medical context and the algorithm's reliability during fallback scenarios.
- **System Performance:** Benchmarking concurrent request handling and latency under simulated multi-hospital traffic loads.

6. Deployment and Handover The final Proof of Concept system is targeted for local deployment within a simulated network environment (such as the Concepcion District Hospital simulation framework) to demonstrate a functional clinic-to-hospital data bridge. Complete system documentation, covering API connection instructions, data dictionaries, deployment guides, and a roadmap for deploying a fully localized, on-premises AI server, will be handed over to WAH management to support long-term sustainability.

7. Maintenance and Feedback Following the Proof of Concept deployment, the development team will collect feedback regarding data translation accuracy, processing efficiency, and system reliability. While long-term operational maintenance falls under WAH management due to the academic scope of a student capstone, the goal is to deliver the updated ADAPT middleware in a modular, highly maintainable, and well-documented state so WAH engineers can easily maintain the system and integrate local AI models in future production environments.

Requirements Analysis

Product Backlog / User Stories

Epic	Story ID	Actor	User Stories	Acceptance Criteria (Given / When / Then)
1: Health Record & Data Interoperability	US1.1	WAH4PCE Admin	As a WAH4PCE Admin, I want to administer the system to securely manage and queue local health records so that they are ready for validation and mapping.	Given a batch of local health records is received, When the system queues the data, Then the admin can view the pending records awaiting mapping.
	US1.2	WAH4PCE Admin	As a WAH4PCE Admin, I want to oversee the interoperability pipeline so that legacy systems can establish a functional exchange with DOH standardized systems.	Given the system is actively processing records, When the admin accesses the pipeline overview, Then they can view real-time metrics of successful transfers, pending queue sizes, and mapping errors.
	US1.3	DOH iHomis / External	As a DOH iHomis or External endpoint, I want to request standardized health data (e.g., HL7 or FHIR 4) so that patient medical history follows them instantly.	Given a patient referral is initiated, When the external system queries the intermediary, Then it successfully receives the full patient history formatted as standard FHIR R4 or HL7 resources.

	US1.4	WAH / WAH4B User	As a WAH or WAH4B facility user (WAH4H/WAH4C), I want to manage and provide health data seamlessly so that I do not have to manually re-encode data during patient referrals.	Given a patient record is transferred via the intermediary, When the receiving staff opens the patient's profile, Then all mapped data populates correctly without requiring manual data entry.
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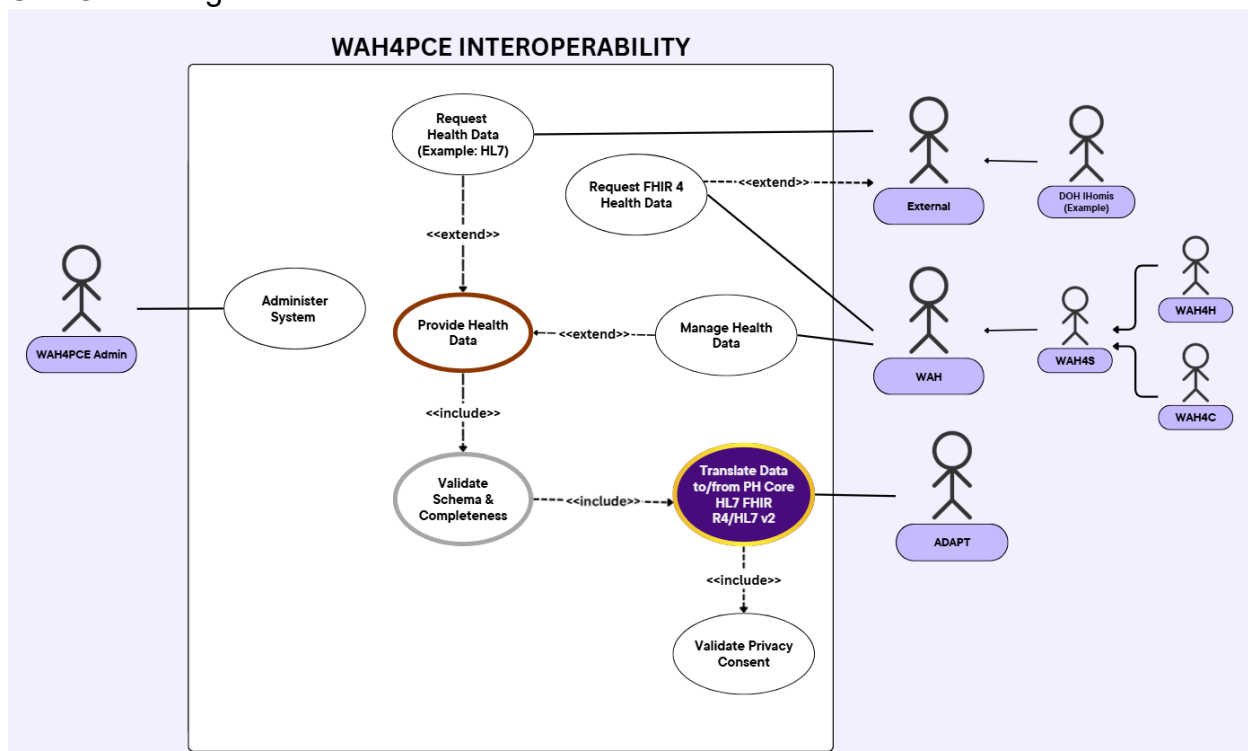
Epic	Story ID	Actor	User Stories	Acceptance Criteria (Given / When / Then)
2: AI-Driven Data Transformation	US2.1	ADAPT	As ADAPT, I want to automatically translate data to and from PH Core HL7 FHIR R4/HL7 v2 standards so that the system removes the need for manual re-encoding.	Given a legacy data payload is routed to the server, When ADAPT processes the text, Then it outputs a payload where all identified fields strictly map to the equivalent PH Core dictionaries.
	US2.2	ADAPT	As ADAPT, I want to transform uniquely formatted local data into standardized models so that disparate healthcare systems can share and interpret data.	Given unstructured local data, When ADAPT transforms it, Then the output categorizes the information correctly into specific resource models (e.g., separating symptoms and vitals).
	US2.3	ADAPT	As ADAPT, I want to translate legacy DOH formats into WAH formats and vice versa so that bi-directional semantic interoperability is achieved.	Given a bi-directional exchange request, When data moves from DOH back to WAH, Then ADAPT accurately translates the

				semantic terminologies so clinical meaning is preserved.
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Epic	Story ID	Actor	User Stories	Acceptance Criteria (Given / When / Then)
3: Automated Verification & Compliance	US3.1	ADAPT	As ADAPT, I want to execute a "Validate Schema & Completeness" check on provided health data so that structural accuracy is verified before translation occurs.	Given health data is provided to the system, When the pipeline initiates, Then ADAPT must pass the payload through a strict schema validation before triggering the translation process.
	US3.2	ADAPT	As the WAH4PCE AI, I want to validate translated resources against the PH Core schema so that structural accuracy is verified before transmission.	Given the AI has finished mapping a record, When the system runs a pre-transmission check, Then the generated resource must pass strict schema validation before delivery.
	US3.3	WAH4PCE Admin	As a WAH4PCE Admin, I want the system to quarantine records that fail the completeness check (missing critical fields) so that I can perform manual reviews and prevent data pollution.	Given a record is undergoing the schema and completeness check, When the system detects a missing mandatory field, Then it redirects the record to a 'Quarantine' queue and flags the admin for intervention.

Epic	Story ID	Actor	User Stories	Acceptance Criteria (Given / When / Then)
4: Middleware Load Management	US4.1	Go Middleware	As the Go middleware orchestrator, I want to automatically switch between AI models when API rate limits are reached so that automated exchange continues without downtime.	Given the orchestrator is routing requests, When the primary AI model returns a 429 Rate Limit error, Then the middleware seamlessly redirects queued payloads to a fallback model.
	US4.2	System Router	As the system router, I want to establish secure TLS connections so that the data payload remains encrypted during model switching and transmission.	Given data is moving between endpoints, When a connection is established, Then the system enforces TLS 1.2+ encryption and terminates unencrypted HTTP attempts.

Use Case Diagram



Use Case Full Description/Fully Dressed Use Case

Use Case Name:	Administer System
Scenario:	The system administrator accesses the interoperability platform to configure endpoints, manage user credentials, and monitor system health.
Triggering Event:	The WAH4PCE Admin navigates to the administrative portal and logs in.
Brief Description:	This use case governs the administrative oversight of the platform, allowing the admin to ensure network uptime, manage security, and configure system parameters.
Actors:	Primary: WAH4PCE Admin
Related Use Cases:	None
Stakeholders:	• WAH4PCE Admin: Needs secure, comprehensive access to system settings. • All Users: Rely on the admin to keep the system operational and secure.
Preconditions:	The WAH4PCE Admin must be authenticated with valid administrative privileges.
Postconditions:	System configurations are successfully updated, applied, and logged in the immutable audit trail.
Flow of Activities:	• Actor: WAH4PCE Admin logs into the system dashboard and selects a configuration module. • System: Authenticates the Admin and presents the requested administrative interface. • Actor: Admin inputs changes or acknowledges system alerts. • System: Commits the changes, updates the system state, and logs the activity.
Exception Conditions:	E1: Invalid Credentials. If the Admin provides incorrect login details, the System denies access and logs a failed attempt.

Use Case Name:	Provide Health Data 41
Scenario:	A core data ingestion or routing event occurs on the platform, acting as the base transaction engine that can optionally be extended by requests, management, or translation processes.
Triggering Event:	Triggered implicitly when an actor initiates a data transaction (either via the Request or Manage extension points).
Brief Description:	This is the foundational data handling use case. It represents the core pipeline for routing health data within the system, automatically including translation protocols when necessary, and serving as the base that other use cases extend.
Actors:	Primary: System (External, WAH)
Related Use Cases:	• Request Health Data (Example: HL7) (<<extend>>) • Manage Health Data (<<extend>>) • Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders:	• External / WAH Actors: Depend on this core pipeline to route data securely. • Patients: Depend on this pipeline to handle their data safely.
Preconditions:	A data transaction must be initiated by an authorized external or WAH actor.
Postconditions:	The data payload is successfully processed, translated (if necessary), and routed to its appropriate destination.
Flow of Activities:	• Actor: (Triggered by extending use cases). • System: Initializes the core data pipeline and evaluates the incoming payload. • System: Hits the <<include>> point to trigger the "Translate Data..." use case to ensure standard formatting. • System: Concludes the transaction and finalizes the routing of the health data.
Exception Conditions:	E1: Pipeline Failure. If the core system experiences a critical error, the transaction is halted, and an alert is sent to the WAH4PCE Admin.

Use Case Name:	Validate Schema & Completeness
Scenario:	Incoming health data is checked for structural integrity, valid consent, and the presence of mandatory fields (e.g., PhilHealth ID) before being allowed to proceed to the translation engine.
Triggering Event:	Automatically triggered by the "Provide Health Data" base use case upon ingestion of a new payload.
Brief Description:	This included use case acts as the "Silver Layer" gatekeeper in the Medallion Data Architecture. It cleanses the raw Bronze payload, ensuring it meets the minimum structural requirements required by the destination endpoint.
Actors:	Primary: System (Internal)
Related Use Cases:	• Provide Health Data (<<include>>) • Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders:	• WAH4PCE Admin: Needs a reliable quarantine log to monitor malformed data. • Patients: Require their records to be complete so clinical care is not delayed.
Preconditions:	A raw data payload (Bronze Layer) must be successfully ingested and staged in the temporary MongoDB database.
Postconditions:	The payload is validated (Silver Layer) and passed to the translation engine, or quarantined if it fails the structural and consent checks.
Flow of Activities:	• System: Ingests the raw payload from the core pipeline. • System: Scans the payload against the predefined schema requirements. • System: Verifies the presence of all mandatory fields and valid privacy consent flags. • System: Marks the payload as validated and passes it down to the Translation step.
Exception Conditions:	E1: Missing Mandatory Field (Data Pollution Risk): If the schema is missing required fields or valid consent, the system halts processing, moves the payload to the Quarantine queue, and alerts the WAH4PCE Admin via the dashboard.

Use Case Name:	Request Health Data (Example: HL7)
Scenario:	An external organization searches the interoperable repository to retrieve specific patient records in legacy or standard HL7 format.
Triggering Event:	An External actor (e.g., DOH iHomis) submits a data query or retrieval request to the platform.
Brief Description:	This use case extends the core data pipeline to allow authorized external entities to pull interoperable health data from the repository in HL7 formats.
Actors:	Primary: External (Generalized actor including DOH iHomis)
Related Use Cases:	Provide Health Data (<<extend>>)
Stakeholders:	External Actors: Need accurate and timely access to patient records.
Preconditions:	The External actor must be authenticated and authorized to access the requested data scope.
Postconditions:	The requested HL7 data is securely delivered to the actor, and the access event is logged.
Flow of Activities:	• Actor: External actor submits a data request query via the integration endpoint. • System: Verifies identity and authorization scope. • System: Executes the core "Provide Health Data" pipeline to retrieve and route the data. • System: Transmits the requested data back to the Actor.
Exception Conditions:	E1: Data Not Found. If no records match, the System returns a '404 Not Found' status.

Use Case Name:	Request FHIR 4 Health Data
Scenario:	A WAH-affiliated entity requests modern, highly structured FHIR R4 clinical data from the interoperability platform.
Triggering Event:	A WAH actor submits an API request specifically for FHIR R4 formatted data.
Brief Description:	This use case governs the secure querying and extraction of modern FHIR R4 health data tailored for actors within the WAH ecosystem, with potential extended access to External actors.
Actors:	Primary: WAH (Generalized actor including WAH4H, WAH4S, WAH4C), External
Related Use Cases:	None directly valid as an extension to UCs; extends actor access.
Stakeholders:	WAH Actors: Need structured, modern data schemas for their specific clinical applications.
Preconditions:	The actor must be authenticated and the requested data must exist in or be translatable to FHIR 4.
Postconditions:	The requested FHIR 4 data is securely delivered to the actor.
Flow of Activities:	• Actor: WAH actor submits a query for FHIR R4 resources. • System: Authenticates the actor and verifies access permissions. • System: Queries the repository for the FHIR resources. • System: Encrypts the payload and transmits it back to the Actor.
Exception Conditions:	E1: Unauthorized Access. If the actor requests resources outside their scope, the system denies the request and logs a security exception.

Use Case Name:	Manage Health Data
Scenario:	An authorized data owner updates, archives, or amends existing health records they previously submitted to the platform.
Triggering Event:	A WAH actor sends an update or modification payload for a specific existing record.
Brief Description:	This use case extends the core data pipeline to allow primary WAH actors to maintain the accuracy and lifecycle of their submitted data, ensuring the ecosystem contains the most current clinical information.
Actors:	Primary: WAH (Generalized actor including WAH4H, WAH4S, WAH4C)
Related Use Cases:	Provide Health Data (<<extend>>)
Stakeholders:	WAH Actors: Require the ability to amend patient information as clinical situations evolve.
Preconditions:	The target record must exist, and the WAH actor must be the authoritative owner of that record.
Postconditions:	The record is successfully updated (with version history maintained), and an audit trail is logged.
Flow of Activities:	• Actor: WAH actor submits an update payload for an existing health record. • System: Validates the Actor's authority to modify this specific data asset. • System: Triggers the core "Provide Health Data" pipeline to process the payload. • System: Applies the updates, versions the record, and sends a success confirmation.
Exception Conditions:	E1: Record Locked. If the record is currently in use or locked by another process, the system rejects the update and prompts a retry.

Use Case Name:	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Scenario:	Data flowing through the core pipeline requires structural normalization to ensure it matches the PH Core HL7 FHIR R4 or HL7 v2 standards.
Triggering Event:	Mandatorily triggered by the "Provide Health Data" use case when data ingestion or extraction occurs.
Brief Description:	This included use case utilizes an AI agent to perform complex semantic mapping and translation of health data schemas, ensuring true interoperability across the platform while enforcing a mandatory privacy check.
Actors:	Primary: WAH4PCE AI
Related Use Cases:	Provide Health Data (<<include>>) Validate Privacy Consent (<<include>>)
Stakeholders:	• WAH4PCE AI: Requires parameters to map data schemas accurately. • All System Users: Depend on this process for standardized data.
Preconditions:	A data payload must be staged in memory, and the AI translation service must be operational.
Postconditions:	The data payload is successfully mapped to standard formats without losing clinical meaning.
Flow of Activities:	• System: Routes the data payload to the AI agent during the core transaction. • Actor: WAH4PCE AI analyzes the schema and performs semantic mapping. • System: Pauses translation to trigger the mandatory "Validate Privacy Consent" use case. • Actor: WAH4PCE AI finalizes the standardized output and returns it to the pipeline.
Exception Conditions:	E1: Translation Failure. If the AI cannot safely map the data without altering clinical meaning, it halts, quarantines the payload, and alerts the admin.

Use Case Name:	Validate Privacy Consent
Scenario:	Before translation and processing can finalize, the system checks the patient's active consent directives to ensure data sharing is legally permitted.
Triggering Event:	Automatically and mandatorily triggered by the "Translate Data..." use case.
Brief Description:	A strict, internal compliance check that cross-references patient identifiers against consent registries to approve or block the processing of health information.
Actors:	Primary: System
Related Use Cases:	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders:	• Patients: Require strict adherence to their privacy choices. • WAH4PCE Admin: Needs a reliable audit trail of compliance checks.
Preconditions:	Patient identifiers must be parsable from the payload, and the consent registry must be online.
Postconditions:	A consent status is appended to the transaction, and the verification event is logged.
Flow of Activities:	• System: Extracts patient identifiers from the active payload. • System: Queries the consent registry for the patient's sharing permissions. • System: Evaluates permissions against the current transaction. • System: Returns an authorization approval back to the translation process.
Exception Conditions:	E1: Consent Revoked. If the registry indicates an opt-out, the System immediately purges the payload, aborts processing, and logs a compliance rejection.

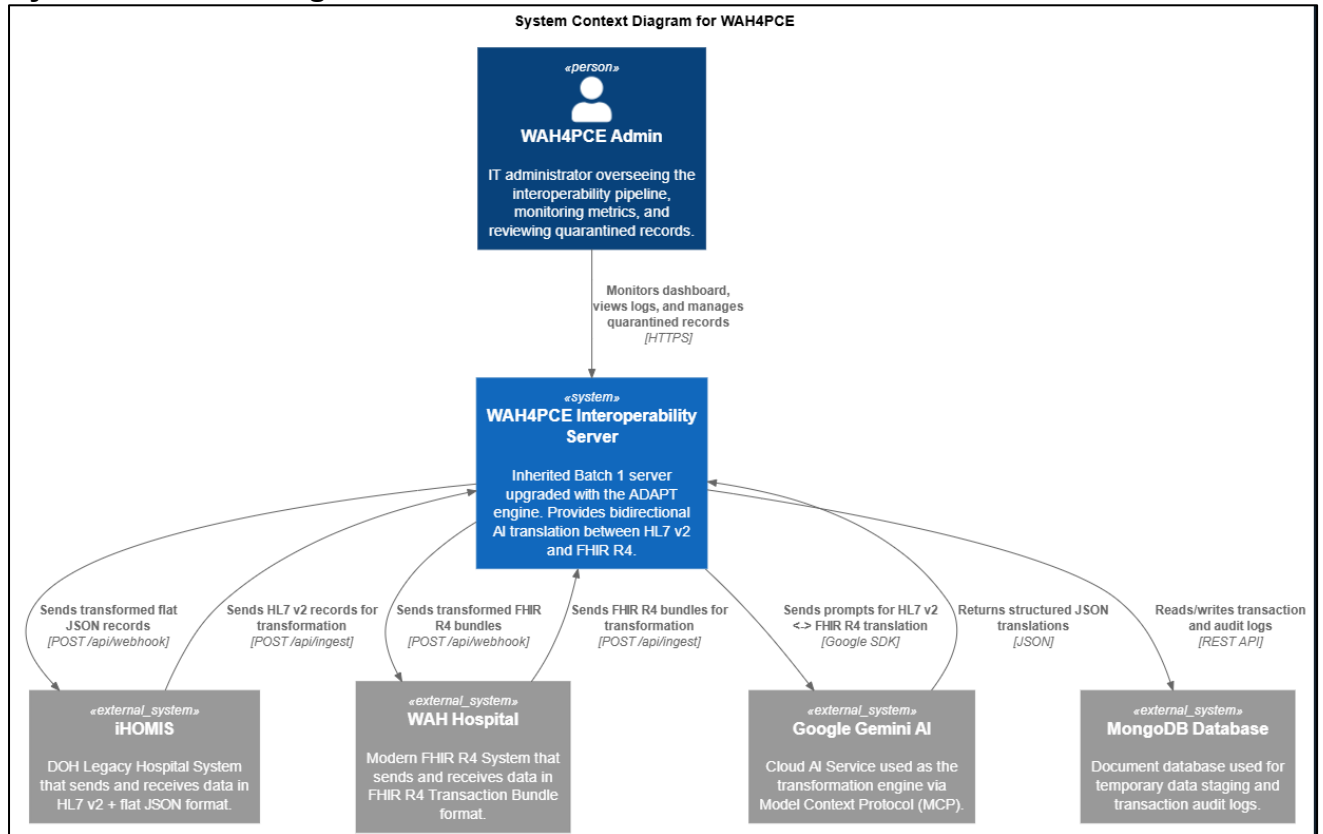
Gap Analysis

Product Backlog ID	Current System	Proposed Changes	Impact
US1.1, US1.2, US1.3, US1.4 (Health Record & Data Interoperability)	Internal-Only Scope & Manual Re-Encoding: The baseline system is restricted to communication within the internal WAH ecosystem. It lacks external connection points for government endpoints, forcing healthcare personnel to manually re-type patient information when referring patients to iHOMIS hospitals.	Standardized Bidirectional API Gateway: Upgrades the platform with external-facing connection endpoints that bridge WAH (FHIR R4) and DOH iHOMIS (HL7 v2) formats.	Enables smooth cross-facility patient referrals, significantly reducing double data entry and administrative delays for healthcare staff.
US2.1, US2.2, US2.3 (Data Transformation & Mapping)	Unchecked Data Flow & Manual Privacy Checks: The system lacks automated filters to catch incomplete forms before transmission, risking data pollution. Verifying patient consent relies on manual review rather than	Medallion Data Validation & Automated Legal Gatekeeper: Integrates structured data filtering (Bronze, Silver, Gold layers) that automatically quarantines incomplete records and blocks any transfer lacking an explicit digital	Proactively prevents corrupted or incomplete records from entering hospital databases and enforces strict compliance with the Data Privacy Act.

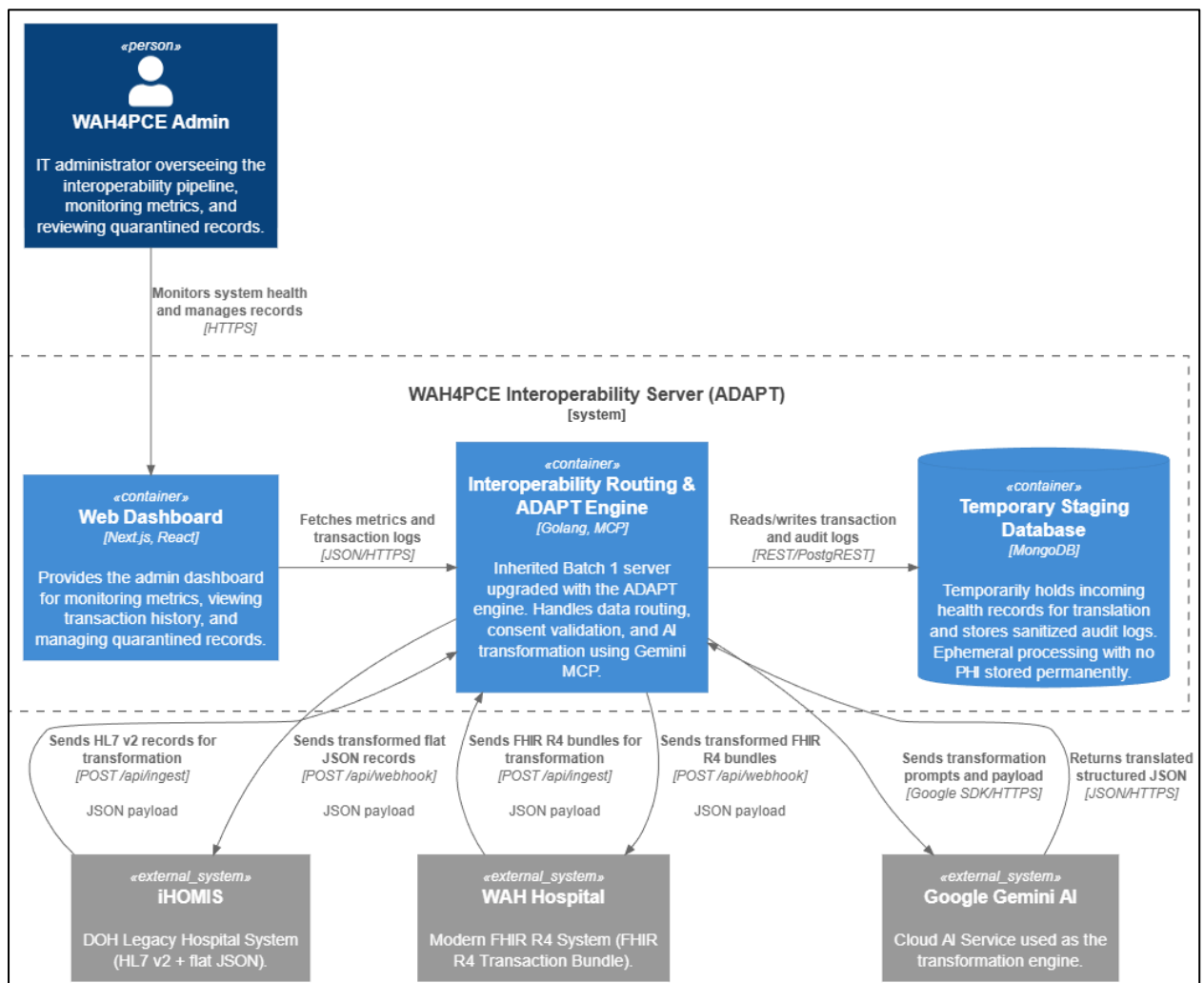
	system enforcement.	patient consent flag.	
US4.1, US4.2 (Traffic Concurrency & Oversight)	Concurrency Limits & Headless Operation: Built as a background Python service without a visual user interface, creating traffic bottlenecks during simultaneous multi-hospital transfers and offering zero visual monitoring for IT staff.	High-Concurrency Engine & Next.js Admin Dashboard: Re-engineers the core routing engine in Go (Golang) for high-speed concurrent processing and adds an interactive web dashboard for real-time traffic monitoring and quarantine management.	Resolves traffic congestion across simultaneous network connections and equips local IT administrators with visual control over pipeline health and quarantined files.

System Analysis and Design

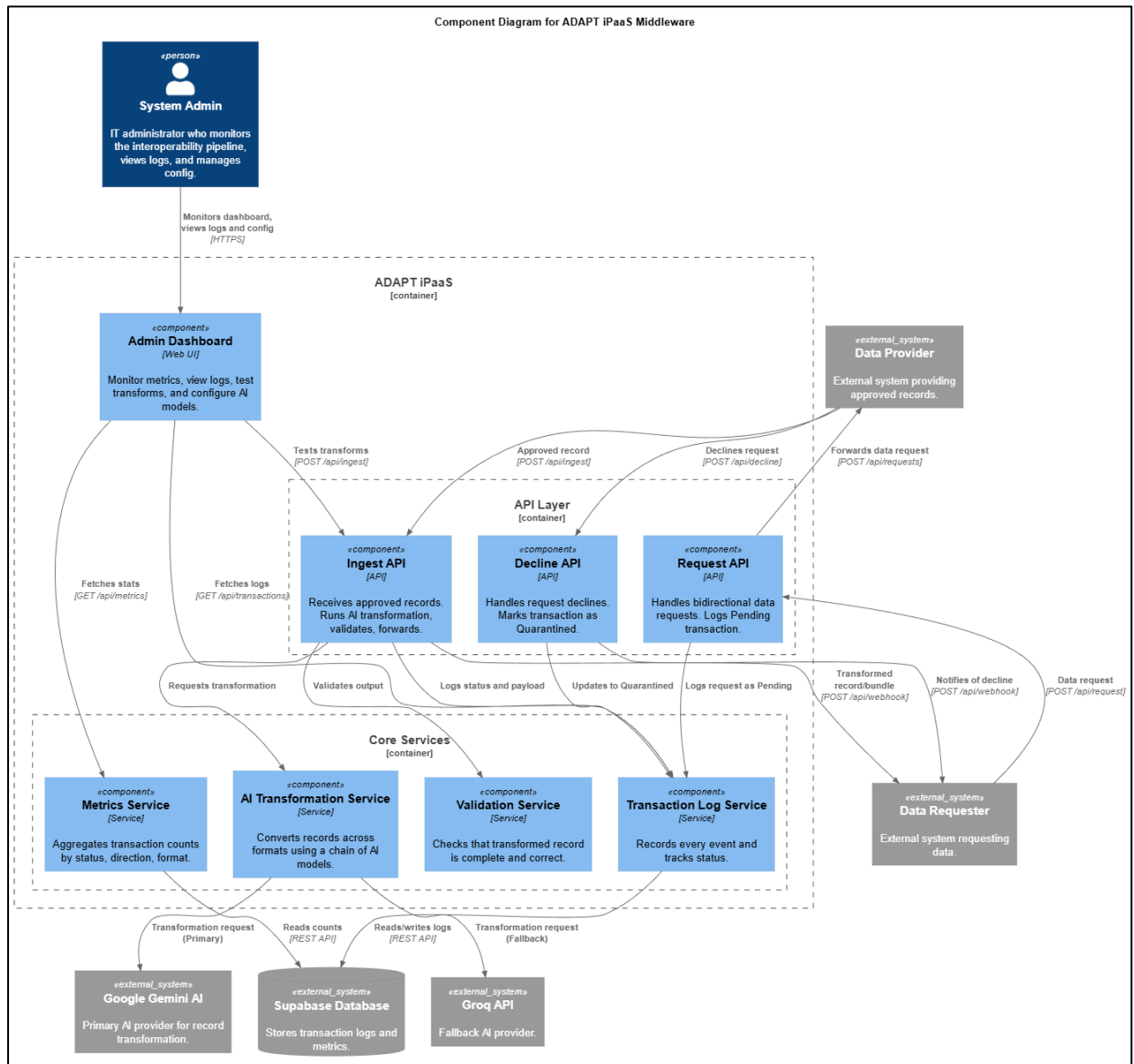
System Context Diagram



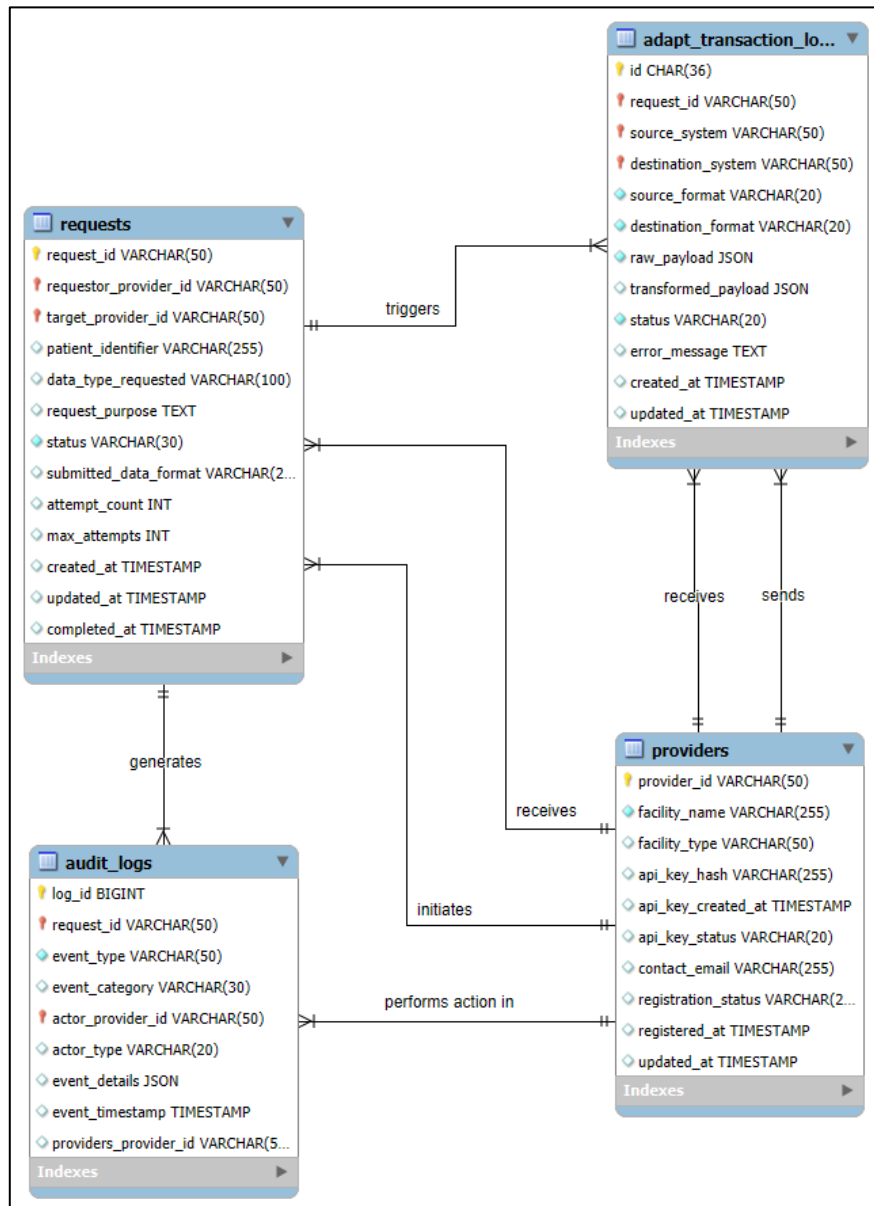
Container Diagram



Component Diagram



Entity Relationship Diagram



I. Results and Discussion

(

Write a short introduction that describes what this section will discuss.

)

5.1 Release Plan

(

Include here your updated release plan and discuss whether you are on/ahead/behind schedule. Explain also how much of the product backlog has been finished (it is up to you how you will present it, without being repetitive. You can say something like, “The complete product backlog is in Table ____”.

)

5.2 Prototype

(

Include here diagrams/illustrations of your proposed solution. Include the latest screenshots. Note that each figure must be properly captioned and discussed in at least one paragraph.

)

5.3 Use Classes and Characteristics

(

Insert a table here with two columns: Roles and Description

<i>Roles</i>	<i>Description</i>
<i>Student Org Representative</i>	<i>A student organization representative is a type of user. When this user makes a request, the request must be approved by the org adviser, etc..</i>
<i>Org adviser</i>	<i>This user approves or disapproves the request made by the representatives of the org he/she handles.</i>

)

II. Conclusion

(

Write what you have accomplished so far (which of your objectives have been achieved and what needs to be done to achieve the others, if there are still any left). Summarize what you have done: specify the objectives and briefly describe how it was achieved. At the end of this chapter, write what will be done for the next iteration (for the next course in the PBL flow of subjects—MCSPROJ)

)

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Appendices

Appendix A: Project Vision

(

Write your updated project vision, please refer to the reference uploaded in MS Teams

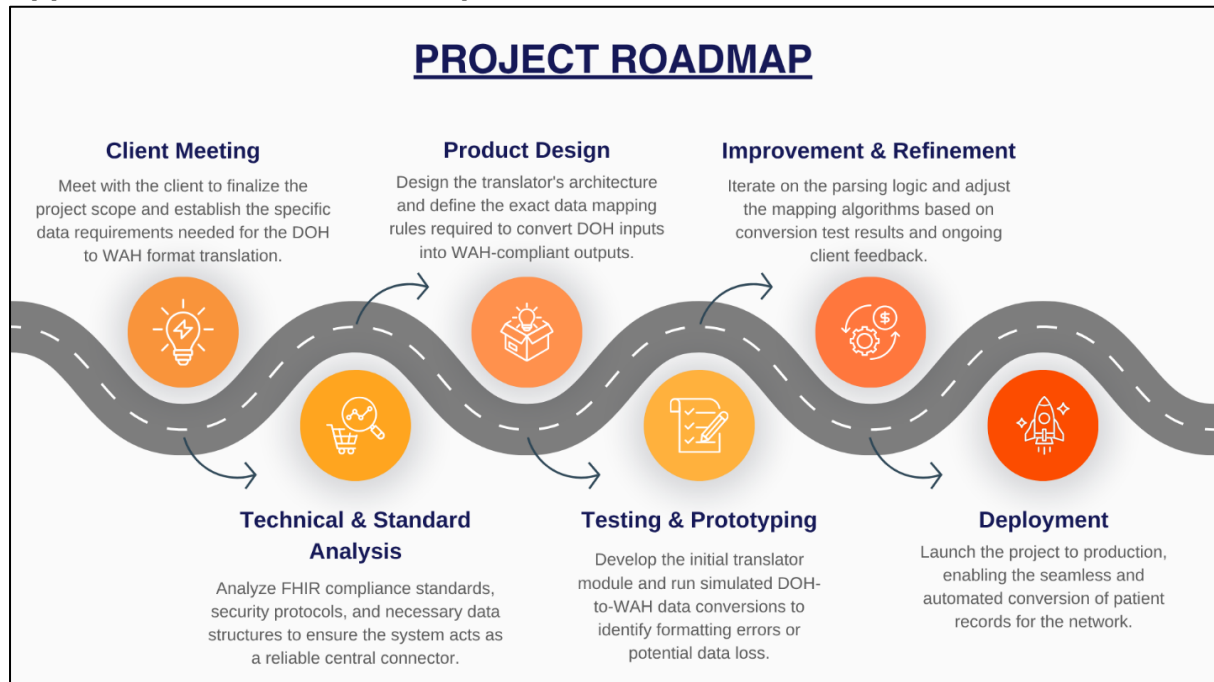
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Appendix B: Gantt Chart



Jira Share Link: <https://student-team-m95p1jfh.atlassian.net/jira/software/projects/MC/boards/2/timeline?rangeMode=WEEKS&atlassianOrigin=eyJpIjoZDMzYjRhNTk2Njk5NGFmYWlzMzdmMWExZWZjZWQ3NzkiLCJwIjoiaSJ9>

Appendix C: Product Roadmap



Appendix D: Users' Manual

(

Add this section only if you think Section 5.2 is not sufficient for users to grasp how your system works.

)

Appendix E: Team Meetings

Meetings with Project Adviser

With Project Adviser - Minutes of the Meeting 1

1. Meeting Details

- **Project / Topic:** Group Merge Conflict Project Advisory Consultation
- **Date:** August 4, 2026
- **Time:** 03:36
- **Platform:** Microsoft Teams
- **Recorded By:** Marwin John Gonzales
- **Attendees Present:**
 - **Gil Ballesca** (Project Adviser)
 - **Rick Francis Cruz** (Backend Developer)
 - **Qelvin Joszeler Nagales** (Project Manager / Lead)
 - **Vince Nelmar Alobin** (Full-Stack Developer)

2. Key Agenda & Discussions

A. Adviser Introduction & Role Alignment

- **Advisor Experience & Background:** Adviser Gil Ballesca shared his background, including working remotely for an AI company established with partners in Australia. He expressed appreciation for being selected as the team's project adviser.
- **Consultation Expectations:** Mr. Ballesca clarified that while he will be advising the team collectively, he will also hold individual consultations with team members based on their specific roles (e.g., backend, full-stack, frontend) to review responsibilities and progress.

B. Technical Stack & Development Status

- **Roles & Responsibilities:**
 - **Rick Francis Cruz:** Serves primarily as the Backend Developer.
 - **Vince Nelmar Alobin:** Serves as the Full-Stack Developer, collaborating with Qelvin on client-side requirements and working closely on the backend and database architecture.
- **Technology Stack:**
 - **Frontend & Framework:** Next.js (built on React), following the tech stack established by previous student batches (e.g., iBugs, TCG Trio).
 - **Database & Staging:** Currently using dummy data / Convex for prototyping and demonstration purposes to transfer data across systems.
- **Current Progress:**
 - **Prototyping:** Although formal code development isn't required yet at this stage, the team has already created a working prototype to help the thesis panel visualize system workflows.
 - **Database Schema:** The team is currently in the **diagramming phase** (mapping schema, relationships, and metadata) and will establish the final database schema ahead of the final defense.

C. Scope Management & Advisory Guidance

- **Addressing Panel Feedback:**
 - Rick noted that while the client (WAH) approved their working prototype, the main challenge during midterms/defense was panel feedback regarding **paper wording and scope**.
 - Panel members for the project were identified as Ms. Roselle, Sir Gonz, and Sir June, alongside client representatives from WAH.
- **Scope Realism & "Over-scoping":**

- o Adviser Gil Ballesca warned against **over-scoping** the project. He noted that academic panels sometimes request overly complex features (e.g., full hospital management systems or complex agentic AI) that exceed realistic student timeframes.
- o He emphasized that the goal of the thesis is to **prove a specific working theory/solution** to a defined problem within a 1-to-2-hour defense, rather than trying to build a multi-billion dollar enterprise application.
- **Strategic Alignment:**
 - o The adviser committed to helping the team structure, negotiate, and defend a balanced, proper scope in their documentation and presentation so panel requirements remain fair, realistic, and achievable.

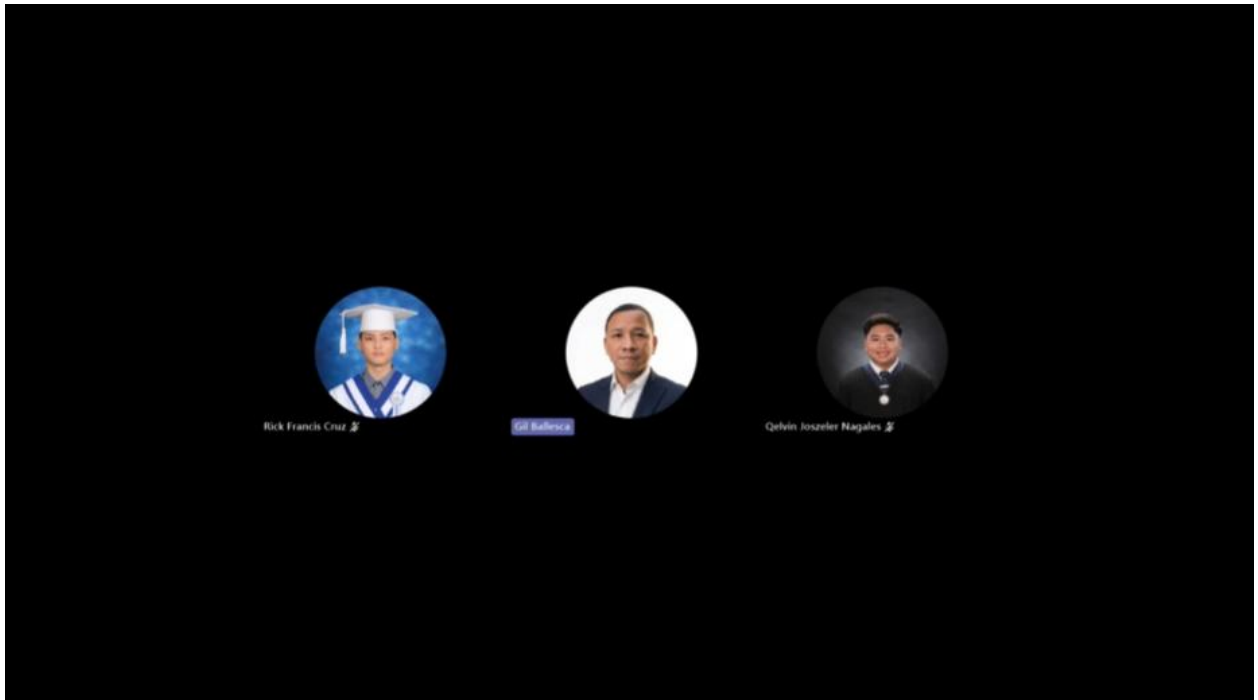
3. Action Items

#	Action Item	Assigned To	Target Timeline
1	Share the compiled panel recommendations and paper draft in the team group chat for adviser review.	Qelvin Nagales & Rick Cruz	Immediate / Within the week
2	Finalize database schema diagrams, entity relationships, and metadata definitions.	Rick Cruz & Vince Alobin	Next Advisory Check
3	Review paper wording and trim down over-scoped objectives to focus strictly on the core problem statement.	Whole Team & Gil Ballesca	Ongoing / Prior to Defense

4. Meeting Adjournment

The meeting concluded with closing remarks and words of encouragement from Adviser Gil Ballesca as the team prepares for their upcoming documentation reviews and in-person alignment.

5. Meeting Documentation



With Project Adviser - Minutes of the Meeting 2

1. Meeting Details

- **Project / Topic:** WAH4PCE Architecture Review & Advisory Consultation (Merge Conflict Team)
- **Date:** August 7, 2026
- **Time:** 3:30 PM – 4:30 PM (PST)
- **Venue:** Asia Pacific College Consultation Room - B 403D
- **Attendees Present:**
 - **Gil Balleza** (Project Adviser)
 - **Marwin John Gonzales** (Project Manager / Developer)
 - **Qelvin Joszeler Nagales** (Team Lead / Project Manager)
 - **Vince Nelmar P. Alobin** (Co-Lead / Developer)
 - **Rick Francis T. Cruz** (Backend Developer / QA)

2. Key Agenda & Discussions

A. Paper Review & Structure Alignment

- **Introduction & Problem Statement:** Reviewed the core background of bridging Wireless Access for Health (WAH) facilities with external iHOMIS systems to overcome disconnected "data islands."
- **Objectives & Scope:** Evaluated the overall scope of the project to ensure the team's goals remain achievable within the academic and development timeline without being overly ambitious or under-scoped.
- **Significance of the Study:** Discussed how the project benefits WAH, DOH, healthcare workers, and patients by reducing double data entry and facilitating smooth record transfers.

B. Specific Adviser Feedback & Technical Direction

- **Data Privacy Requirements (Client - WAH):**
 - **Guidance:** Adviser Gil Ballesca emphasized that the team must conduct deeper research into the client's (WAH) data privacy requirements and legal constraints under the Data Privacy Act (RA 10173).
 - **Action:** Further investigate automated consent flags and waiver handling to ensure compliance before any cross-facility data exchange occurs.
- **Architecture Scope (Direct Hospital-to-Hospital vs. Centralized Database):**
 - **Guidance:** The adviser noted that establishing a full direct hospital-to-hospital (p2p/federated) routing network in one of the current objectives is too large and overly ambitious for the current project scope.

- o **Direction:** The team was advised to **maintain/keep a centralized database** approach to ensure the system architecture remains manageable, stable, and deliverable.
- **API Key Security & Storage:**
 - o **Guidance:** The adviser raised concerns regarding key management and security.
 - o **Direction:** The team needs to put more thought into safe storage strategies for API keys (e.g., environment variables, secure key vaults, or encrypted server storage) to prevent unauthorized access.
- **Use Case Diagram (UCD) Revision:**
 - o **Guidance:** The adviser reviewed the Use Case Diagram in the paper and noted that the current version is **too simple** and lacks the necessary depth to reflect the actual system capabilities.
 - o **Direction:** The UCD needs to be expanded and updated to clearly define actors, system boundaries, and sub-processes (such as validation, consent checking, and quarantine queues).

3. Action Items

#	Action Item	Assigned To	Target Timeline
1	Research client-specific (WAH) data privacy protocols and update the consent validation workflow in the documentation.	Vince Alobin & Rick Cruz	Next Advisory Check

2	Revise the project objectives and architecture diagrams to align with a centralized database structure instead of direct hospital-to-hospital routing.	Marwin Gonzales & Vince Alobin	Next Advisory Check
3	Formulate a secure API key storage strategy and document it in the Technical Background section.	Rick Cruz	Next Advisory Check
4	Redraw and expand the Use Case Diagram (UCD) to add proper operational detail and complexity.	Qelvin Nagales & Vince Alobin	Next Advisory Check

4. Meeting Adjournment

The consultation ended at 4:30 PM with clear directives on refining the project paper and scaling back ambitious objectives into a solid, defensible thesis scope.

5. Meeting Documentation



With Project Adviser - Minutes of the Meeting 3

Meeting Details

- **Meeting Title:** Merge Conflict – Weekly Meeting
- **Date & Time:** August 14, 2026 | 3:30 PM – 4:30 PM (UTC 07:30)
- **Organizer / Recorder:** Qelvin Joszeler Nagales
- **Platform:** Microsoft Teams

Attendees

- **Advisor / Mentor:** Gil Ballesca
- **Team Members:**
 - Qelvin Joszeler Nagales
 - Vince Nelmar Alobin
 - Rick Francis Cruz
 - Marwin John Gonzales

Agenda & Discussion Highlights

1. Project Updates & Previous Meeting Follow-ups

- **Sync with WAH:** The team reported holding a meeting with WAH the previous day. Key architectural points discussed included the Medallion architecture and Open API considerations.
- **Integration Approaches:** Three approaches were presented to WAH:
 - Pure Algorithm-based
 - Pure AI-based
 - Hybrid (Algorithm + AI Integration)

WAH expressed a preference for whichever approach is most accurate, precise, efficient, and fast for translating health data.

2. Standards & Architecture (HL7 vs. FHIR / PH Core IG)

- **Standard Compatibility:**
 - The team presented the **WAH4PCE Interoperability Use Case Diagram (UCD)**, which was previously checked and approved by their professor (Sir Shan).
 - The discussion covered how external systems (e.g., DOH iHOMIS) use **HL7 v2** while modern frameworks utilize **HL7 FHIR (PH Core IG)**.
 - Sir Gil noted that since the Philippine FHIR standard (Draft PH Core IG) is still under active development and not final, relying too heavily on an unstable specification poses a risk of rework.
 - **Advice on Standards:** The team was advised to anchor core transformation reliability on stable standards (like HL7) and implement proper mapping/adaptor design patterns to manage data variations.

3. Demo & Technical Prototype (ADAPT iPaaS)

- Vince presented the live prototype demonstrating data mapping and transformation between **WAH (FHIR R4)** and **DOH iHOMIS (HL7 v2)**:
 - **Data Privacy Validation:** Showcased automatic validation where records missing patient privacy consent under Republic Act 10173 are quarantined and rejected from being pushed to destination endpoints.
 - **Field Discrepancies & Unmapped Data:** Discussed how iHOMIS contains specific fields (e.g., separate chief complaints, diagnosis descriptions) that may not map 1:1 with standard FHIR conditions.
 - **Recommendation on Custom Fields:** Sir Gil suggested implementing custom fields/metadata handling for unmapped attributes to prevent loss of critical clinical context.

4. AI vs. Algorithm Implementation Strategy

- **Token Costs & Sustainability:** Sir Gil emphasized the high recurring costs, token limitations, latency, and potential downtimes associated with relying purely on external LLMs/AI APIs for high-volume enterprise health data.
- **Recommended Strategy:**
 - Prioritize a **deterministic, precise algorithm** as the primary engine for schema transformations and data mapping.
 - Use **AI as an auxiliary feature or fallback**, or apply AI to higher-level clinical decision support/triage (e.g., analyzing vital signs anomalies and alerting emergency cases) rather than basic structural field mapping.

Action Items

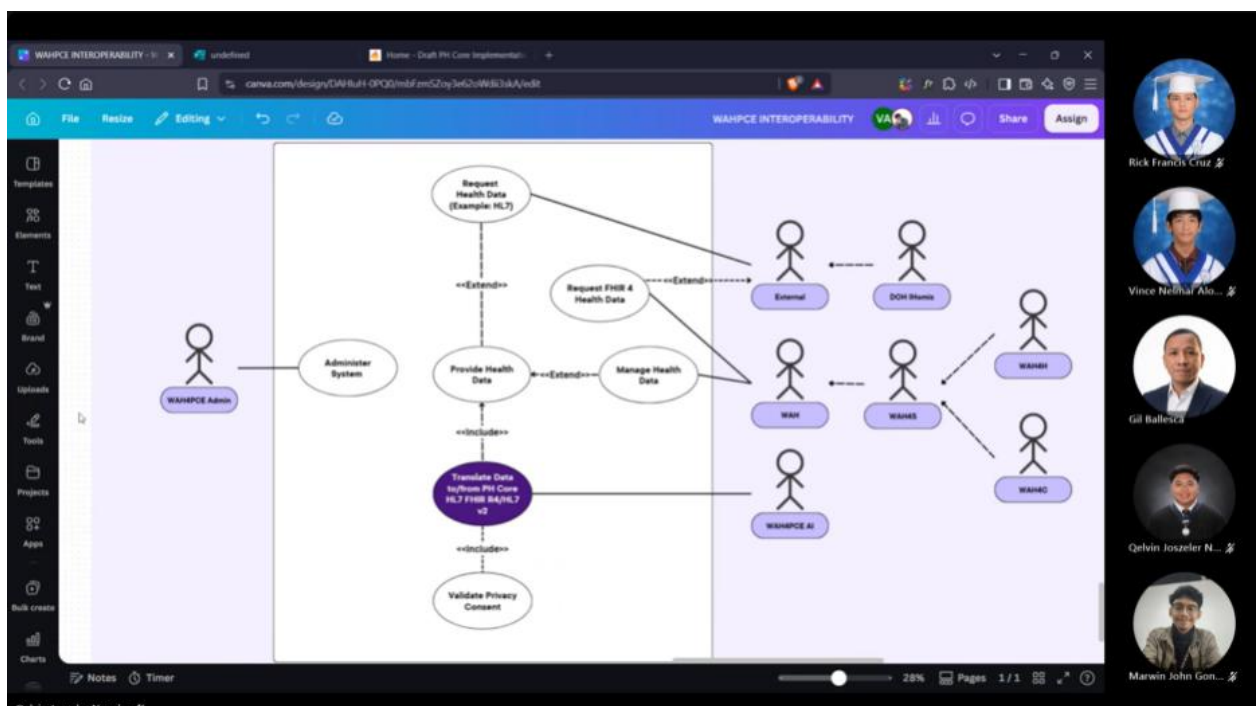
Task / Deliverable	Assignee(s)	Target Date
Update and refine the Use Case Diagram (UCD) to align with the dual Algorithm + AI fallback architecture	Merge Conflict Team	Prior to Project Defense
Formulate algorithm-based mapping logic as the primary pipeline and prepare AI fallback error handling	Vince Nelmar Alobin / Dev Team	Ongoing
Prepare a business proposal/comparison analyzing model costs (Claude, Gemini, GPT) vs. algorithmic processing for WAH	Merge Conflict Team	Next Sync

Update WAH regarding project progress, architecture decisions, and scope boundaries	Merge Conflict Team	Immediate
Add Sir Gil Balleca to the official communication group (Messenger/Email) to ensure he stays in the loop	Qelvin Joszeler Nagales	Immediate

Next Meeting Schedule

- **Date:** Monday, August 17, 2026
- **Time:** 11:30 AM – 12:30 PM
- **Cadence:** Twice weekly (Mondays and Fridays) leading up to project defense

5. Meeting Documentation



With Project Adviser - Minutes of the Meeting 4

Meeting Details

- **Date:** August 18, 2026
- **Time:** 11:29AM PST
- **Platform:** Microsoft Teams

Attendees

- **Gil Ballesca** – Project Adviser
- **Marwin John Gonzales**
- **Qelvin Joszeler Nagales**
- **Rick Francis Cruz**
- **Vince Nelmar Alobin**

Agenda

Review and refine the WAH4PCE Project Documentation (Paper v1.docx), specifically focusing on the Statement of the Problem, Objectives, System Architecture, and Diagram synchronizations in preparation for the upcoming system defense.

Key Discussion Points

1. Algorithmic Engine vs. AI Translation (Fallback Mechanism)

- **Core Shift:** The project title and objectives previously leaned too heavily on being "AI-driven," conflicting with primary requirements for speed, determinism, and cost efficiency.
- **Resolution:** The primary translation engine remains algorithmic (rule-based). A localized AI model will strictly serve as a fallback mechanism for edge cases, unusual data structures, or missing schemas.
- **Audit Logging:** Implemented an audit trail mechanism to tag and flag records processed via AI fallback, alerting receiving systems for manual verification.

2. Documentation & System Architecture Alignment

- **Synchronization:** Ensure written objectives strictly match the system architecture and sequence diagrams, keeping descriptions concise and focused on core data ingestion and interoperability transport.
- **Terminology Cleanup:** Eliminate unintended cloud terminology (e.g., "OpenAPI") to emphasize the fully localized, privacy-compliant AI deployment model.

3. Secure Gateway & Protocol Specifications

- Replace high-level security claims with concrete technical implementations (e.g., TLS handshakes, SSL certificates, payload encryption) across all external interoperability endpoints (DOH, iHOMIS, VGH, OsMak).

4. Data Cleaning, Validation & Medallion Pipeline

- **Layer Definitions:**
 - **Bronze:** Raw data ingestion.
 - **Silver:** Deduplication, schema normalization, structural sanitization, and handling foreign/tourist records.
 - **Gold:** Standardized, interoperable FHIR-ready datasets.
- **Standardization:** Enforce consistent English data dictionaries and field mappings rather than mixed localization tags.

5. AI Reliability & Consent Validation

- Include benchmark documentation showing the 100-run test results to prove zero-hallucination thresholds.
- Implement explicit verification of signed consent forms/documents rather than a simple boolean flag.

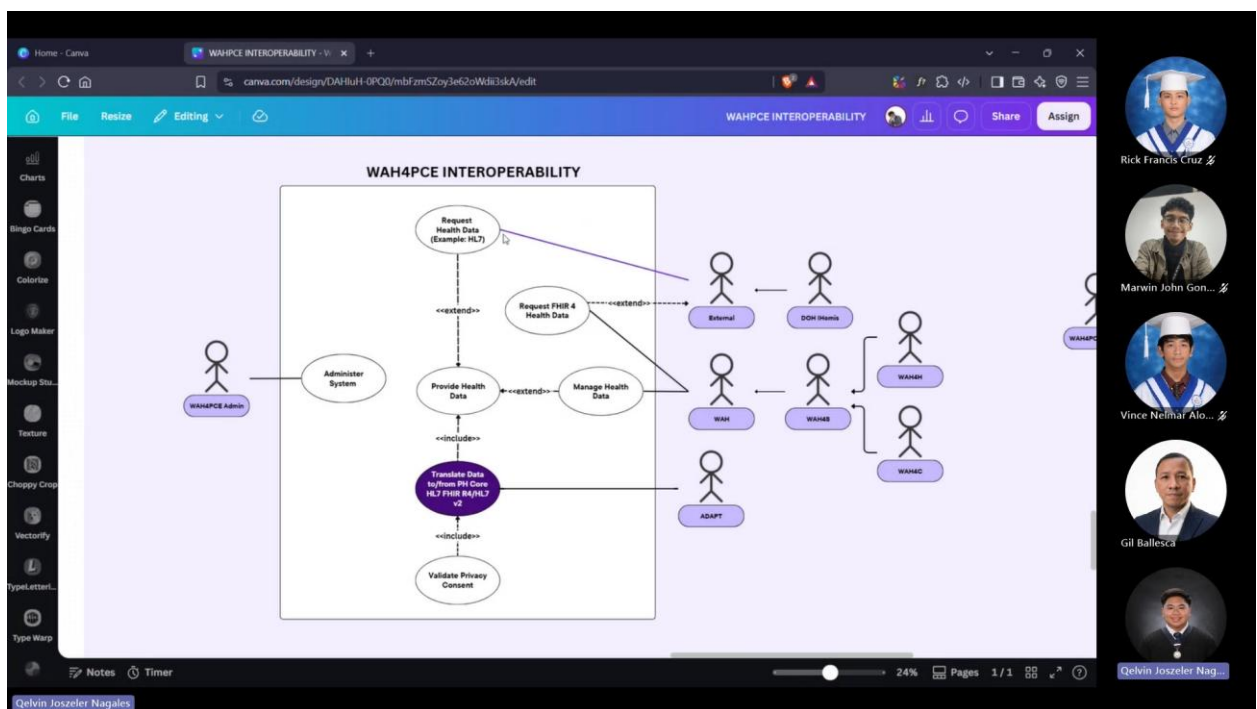
6. Diagram Revisions (ADAPT Engine & UCD)

- Update Use Case Diagrams (UCD) and Architecture Diagrams to illustrate the ADAPT engine decision tree (algorithmic path vs. AI fallback split).

7. Defense Logistics

- Schedule:** August 19, 2026, at 8:00 AM PST
- Panelists:** Sir Gonzalo, Sir Bueno

8. Meeting Documentation



Action Items

Action Item	Assignee	Status
Reframe documentation to set Algorithmic Engine as primary and AI as fallback	Merge Conflict Team	Pending

Remove cloud references ("OpenAPI") and specify localized AI infrastructure	Merge Conflict Team	Pending
Define specific gateway transport security protocols (TLS/SSL encryption)	Merge Conflict Team	Pending
Standardize Medallion transformations (Bronze/Silver/Gold) and data dictionary schemas	Merge Conflict Team	Pending
Update Use Case and Sequence Diagrams with the ADAPT engine workflow	Merge Conflict Team	Pending
Send finalized diagram exports to Sir Gil Ballesca for pre-defense review	Merge Conflict Team	Pending

Meetings with Client

With Client - Minutes of the Meeting 1

1. Meeting Details

- **Project / Topic:** Interoperability Architecture & Data Translation (WAH, iHOMIS, DOH)
- **Date:** August 13, 2026
- **Time:** 11:05 AM – 11:46 AM (PST)
- **Platform:** Google Meet (hiw-rclk-mwr)
- **Attendees Present:**
 - Nicky Balderosa
 - Qelvin Nagales
 - Vince Nelmar Alobin

- o Rick Cruz
- o Robert Michael Martinez
- o Marwin John Gonzales

2. Key Agenda & Discussions

A. System Scope & Deliverables

- **External Connections & APIs:** The team clarified that their scope is to build the intermediary interoperability layer (API endpoints and documentation) so external systems (such as WAH and iHOMIS) can communicate.
- **DOH Reports:** DOH health reports (e.g., PDFs generated via EMR/iHOMIS) are currently extracted/printed and submitted manually. Automating direct report sync from the interoperability layer to DOH is **out of scope**.
- **Midterm Defense Requirement:** For the upcoming midterm defense, the team only needs to present system architecture/diagrams (working code is not required yet).

B. Translation Approach: Algorithm vs. AI

The team discussed the translation of incoming health data into standard JSON formats:

- **Algorithm-Based Translation:** Faster, highly efficient, precise, and cost-effective (no token costs).
- **AI-Based Translation:** Flexible, requires less initial development logic, but incurs API token costs and potential latency.
- **Hybrid / Backup Approach:** Suggestion to primary use algorithm-based translation, with AI serving as a fallback/safety net for edge cases.
- **Evaluation Criteria:** Nicky emphasized that **accuracy and precision** are priority #1. Priority #2 is **maintainability**—the chosen approach must be easy for the core team to maintain long-term.

C. Data Flow & Integration Alignments

- **Data Format Handling:** Vince raised a question regarding JSON conversions for data originating from WAH-Hospital and WAH-Clinic sent to iHOMIS or central storage.
- **Platform Alignment (OpenHIM):** It was noted that other groups (e.g., Fama's team) are looking into **OpenHIM**. Nicky advised coordinating with team leads (Gelo & Sophia) to prevent redundant work and ensure alignment across all 4 sub-teams.

D. Security & Testing

- System security testing and vulnerability enhancements will be handled by the dedicated security sub-team.

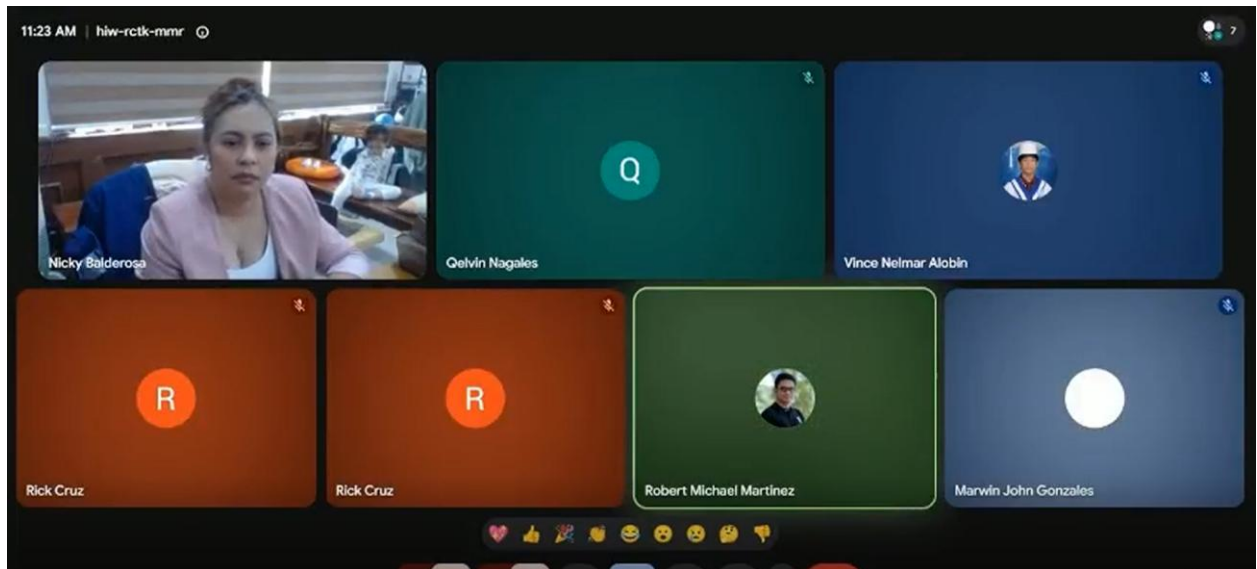
3. Action Items

#	Action Item	Assigned To	Target Timeline
1	Meet with sub-team leads (Jello & Sophia) to align on OpenHIM and avoid redundant effort.	Core Interop Team	Prior to Midterm Defense
2	Finalize architecture diagrams and data flow documentation for the midterm defense.	Core Interop Team	Next Week
3	Evaluate algorithm vs. AI translation trade-offs and select the official primary pipeline.	Core Interop Team	Post-Midterms

4. Meeting Adjournment

The meeting was formally adjourned after addressing all open questions from the participants.

5. Meeting Documentation



Appendix F: Source Code

<https://github.com/APC-SoCIT/APC-2026-2027-T1-IT241-Grp3-WAH4PCE-An-AI-Driven-Interoperability-Bridge-for-iHOMIS>

APC-SoCIT / APC-2026-2027-T1-IT241-Grp3-WAH4PCE-An-AI-Driven-Interoperability-Bridge-for-iHOMIS

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Files

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CSPROJ

INTSDEV

MSYADD1

ERD

Test_Case

component_diagram

container_diagram

system_context_diagram

use_case_diagram

use_case_full_description

readme.md

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README.md

APC-2026-2027-T1-IT241-Grp3-WAH4PCE-An-AI-Driven-Interoperability-Bridge-for-iHOMIS / docs / MSYADD1

Add file

Vince0028 Merge pull request #34 from APC-SoCIT/feat/versions 33e1764 · 4 minutes ago History

Name	Last commit message	Last commit date
..		
ERD	Refactor code structure for improved readability and maintainability	5 minutes ago
Test_Case	moved the Test case into a folder to make it more organized	2 days ago
component_diagram	Refactor ADAPT iPaaS component diagram for improved structure and cla...	last week
container_diagram	Refactor code structure for improved readability and maintainability	2 weeks ago
system_context_diagram	docs: add first revision of the ADAPT iPaaS system context diagram	last week
use_case_diagram	feat: add 3rd revision of the use case diagram to documentation	3 days ago
use_case_full_description	Added the 3rd revision of the Use Case diagram full description	2 days ago
readme.md	new	last month

readme.md

Documentation folder for MSYADD1